Resolving Relatedness in a Reconstructed Lexicon: A Probabilistic Method for Proto-Language Internal Reconstruction

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Abstract

The standard method for establishing the relatedness of linguistic forms in historical linguistics requires evidence from written texts. Where such evidence is unavailable, as in the case of reconstructed proto-languages, relatedness claims are often considered unresolvable. This paper proposes a probabilistic method for determining whether phonemic similarity between two forms in a closed system is likely to be coincidental. Using the Proto-Indo-European phonemic inventory as reconstructed by Beekes (1995) and others, we demonstrate that the three-phoneme root *$h_1$és-, shared by the PIE copula and the third person pronoun paradigm, has a probability of coincidental identity of approximately 1 in 27,000 under uniform distribution assumptions, and approximately 1 in 13,000 under frequency-adjusted estimates that account for the observed rarity of *h_1 in root-initial position, the dominance of *e as the default PIE ablaut vowel, and the frequency of *s in root-final position. This result is shown to be independent of which standard reconstruction of PIE is adopted. We further argue that pronouns and copulas, as core vocabulary items, are among the least likely morphemes to be borrowed, strengthening the assumption that PIE constitutes a closed system for purposes of this calculation.

Evidence from six daughter languages — Sanskrit, Greek, Avestan, Latin, Hittite, and Old Irish — confirms the persistence of the *$h_1$és- root across both the copula and the pronominal system in each branch, surfacing in

full where an ending follows and reducing to *h_1e- only in the bare nominative, where final *-s is regularly lost. Hittite provides direct attestation of the laryngeal rather than requiring its reconstruction from secondary effects. The result is further supported by the observation that languages developing a two-be system — including Old Irish and Spanish — consistently assign the *$h_1$és- reflex to the equational/identificational slot, suggesting a stable cognitive association between this root and pronominal predication across time and geography.

The cross-linguistic implications are explored through comparison with Biblical Hebrew, where an independent pronoun-copula root identity has been previously demonstrated (Katz 1996), suggesting the cognitive alignment of pronominal reference with predicational being is not a PIE-specific phenomenon but a recurring feature of human linguistic organization. The methodological contribution — probabilistic synchronic analysis as a tool for internal reconstruction in closed systems — is applicable beyond PIE to any proto-language where phonemic inventory and core vocabulary can be reliably established.

Section 1. Introduction

When the author observed to a colleague that the PIE copula root *$h_1$és- and the demonstrative pronoun root *$h_1$és-[1] appeared to share a common element, the colleague's response was characteristic: "Probably — but how would you prove it?" The answer, it turns out, is embedded in the question. Probable and prove are doublets, both descending from Latin probare, to test or establish as good. To ask how one would prove a historical connection is already to ask how probable that connection is under the best available evidence. There is no other kind of proof in historical inquiry.

This paper takes that exchange as its methodological starting point. Historical linguistics has always relied on probabilistic reasoning — sound correspondences are statistical regularities, reconstruction is inference to best explanation, and the comparative method as a whole is a structured procedure for identifying non-random patterning across related systems. Yet the field has been reluctant to make this probabilistic foundation explicit, particularly when evaluating proposed internal connections within the reconstructed PIE lexicon. The result is an asymmetry: probability operates silently as the justificatory framework while being rhetorically disavowed as insufficient proof.

What follows is an attempt to make the reasoning explicit.

This paper develops a probabilistic method for determining whether phonemic similarity between two forms in a closed system is likely to be coincidental, and applies it to a specific case: the apparent identity between the Proto-Indo-European copula root *$h_1$és- and the third-person pronoun root, reconstructed as the same *$h_1$és-, surfacing as *h_1e- only in the bare nominative, where final *-s is regularly lost in absolute final position. Using the PIE phonemic inventory as reconstructed by Beekes

*1. The root surfaces as *$h_1$es- throughout the oblique pronoun paradigm (genitive, dative, ablative, locative) and reduces to *h_1e- only in the bare nominative, where final *-s is regularly lost in absolute final position. See Section 5.1.*

(1995), and confirming the result across the major competing reconstructions, the paper shows that the probability of this three-phoneme identity arising by chance is approximately 1 in 27,000 — a figure that remains conservative even under the most restrictive non-laryngealist reconstruction available.

The argument advanced here is not that two independently existing items happen, by coincidence, to be historically related. It is narrower and more specific: that the standard practice of listing the copula and the third-person pronoun as two separate reconstructed morphemes is not supported by the phonemic evidence, and that the probability of their observed identity arising by chance is too low to sustain the paradigmatic distinction as currently maintained. The question is not whether to connect two independent entities, but whether the analytical separation between them was ever justified in the first place.

The case for this claim rests on four converging lines of evidence. The identity is directly attested within the PIE reconstruction itself, in the oblique forms of the third-person pronoun paradigm, without any need to appeal to daughter-language reflexes. It persists, independently, across six attested branches of the family, including Hittite, where the relevant laryngeal is read directly from the cuneiform record rather than reconstructed from secondary effects. It resurfaces again in an unrelated grammaticalization event — the medieval Ibero-Romance development of the Spanish ser/estar distinction — with no possible channel of historical transmission from the original PIE pattern. And a structurally similar pronoun-copula identity has been independently established for Biblical Hebrew, a language unrelated to PIE by any accepted genealogical account, suggesting that whatever produced the pattern in one family cannot be dismissed as a peculiarity of the other.

This is not, it turns out, an entirely new proposal. A pronominal origin for *$h_1$és- was raised before and met with a formal objection concerning the morphological segmentation of the pronoun paradigm — an objection this paper addresses directly in Section 8, arguing that it answers a different question than the one the probabilistic method poses. The present contribution is to show that the question can be settled independently of that earlier dispute, using a method that takes the phonemic evidence itself, rather than its morphological packaging, as the relevant object of analysis.

The paper proceeds as follows. Section 2 establishes the closed-system assumption on which the probability calculation depends. Section 3 establishes the PIE phonemic inventory across four major reconstructions. Section 4 develops the probability calculation itself. Section 5 presents the paradigm evidence, branch by branch, including the PIE-internal evidence that anchors the entire argument. Section 6 addresses the turtles-all-the-way-down objection. Section 7 demonstrates that the result is independent of which reconstruction is adopted. Section 8 engages the existing literature, including the prior proposal and its rejection. Section 9 considers the implications, including the question of whether PIE speakers themselves would have perceived this identity. Section 10 concludes.

Section 2. The Closed System Assumption

The probability calculation developed in Section 4 depends on a second prerequisite alongside the phonemic inventory established in Section 3: the assumption that PIE constitutes a closed system for the elements under comparison. This section defines what that assumption requires, argues that it holds for the copula and pronoun paradigms specifically, and addresses the most obvious objection to it.

2.1 What a Closed System Requires

A probability calculation of the kind developed in this paper asks: given a finite inventory of phonemic contrasts, what is the probability that two forms would share their phonemic content by chance? This question only makes sense if the inventory in question is genuinely the relevant universe from which both forms were drawn. If either form could in principle have been borrowed from an external source — a language outside the system — then the calculation no longer answers the right question. A borrowed form's phonemic shape reflects the phonotactics of its source language, not the probability structure of the system into which it was borrowed.

For the calculation in Section 4 to be valid, then, the copula root and the pronoun stem must both be inherited material — items that arose from within the PIE phonemic system itself, not items introduced from outside it. This is the sense in which PIE must be treated as closed.

2.2 Is PIE a Closed System?

PIE as a whole is not, of course, hermetically sealed. Reconstructed PIE shows contact effects, and the broader prehistory of the language undoubtedly involved contact with neighboring language families. In this general sense, no natural language is a closed system.

But the closed system assumption required here is narrower and more defensible. It does not require that PIE as a whole be free of all external influence. It requires only that the specific items under comparison — the copula and the third-person pronoun — are not loanwords. If these two items are inherited core vocabulary, then whatever contact effects exist elsewhere in the lexicon are irrelevant to the calculation.

2.3 The Borrowing Objection

A reviewer might object that the closed system assumption is doing too much work: how can we know that the copula or the pronoun stem was not borrowed at some point in PIE's prehistory, given that PIE itself is a reconstruction with an unknown and unknowable contact history?

The response to this objection rests on a well-established typological generalization, now supported by large-scale comparative data: certain categories of vocabulary are systematically resistant to borrowing, and pronouns and the copula are among the most resistant categories of all.

2.4 Pronouns and Copulas as Borrowing-Resistant Vocabulary

The Loanword Typology Project (Haspelmath and Tadmor 2009), based on a comparative database of 1,460 meanings across 41 languages, provides the relevant empirical evidence. Among its central findings is that lexical meanings are borrowed more readily than grammatical meanings, and that within the grammatical domain, demonstratives, personal pronouns, interrogatives, and basic polysemous verbs are among the least borrowable categories identified by the project.

This finding is directly relevant to the present argument. Both elements under comparison — the third-person pronoun stem and the copula *$h_1$és- — fall within the categories the project identifies as most resistant to borrowing: a personal/demonstrative pronoun and a basic verb of being.

One qualification is necessary. The same body of research notes that no category is entirely immune to borrowing; even pronoun paradigms can in principle admit loanwords, particularly under conditions of intense and prolonged contact. The claim here is therefore probabilistic rather than absolute: borrowing into these categories is rare, and a closed-system assumption restricted to them is far stronger than one applied to the lexicon as a whole, such as nouns for cultural artifacts or technical innovations, where borrowing is common.

2.5 The Assumption as Applied to *$h_1$és-

Given this, the closed system assumption as applied to the present argument is narrow and well-supported: whatever PIE’s broader contact history may have been, the copula root *$h_1$és- and the third-person pronoun stem sharing its phonemic content belong to exactly the categories that comparative borrowability research identifies as most resistant to external replacement.

This narrows the closed system assumption to its defensible core. The calculation in Section 4 does not require that PIE as a whole be free of contact effects — only that these two specific items, by virtue of their grammatical category, are overwhelmingly likely to be inherited rather than borrowed. On that narrower claim, the assumption is well-grounded by comparative typological evidence rather than by stipulation.

Section 3. The PIE Phonemic Inventory

The probability calculation developed in Section 4 depends on one empirical prerequisite: a reliable count of the phonemic contrasts available in PIE. This section establishes that count across four major reconstructions and demonstrates that the result of the calculation is independent of which reconstruction is adopted.

3.1 The Standard Inventory: Beekes (1995)

The reconstruction of Beekes (1995) serves as the primary reference for this paper. Beekes posits a consonantal inventory of 25 segments, organized as follows.

Category	Segments
Stops: plain voiceless	p, t, k, k̂, k^w
Stops: plain voiced	b, d, g, ĝ, g^w
Stops: voiced aspirate	bh, dh, gh, ĝh, g^wh
Fricative	s
Laryngeals	h_1, h_2, h_3

Category	Segments
Resonants	m, n, r, l, y, w

To these 25 consonants are added 5 phonemic vowels: e, o, a, i, u. Long vowels are treated as positional variants rather than independent phonemes for purposes of this calculation, following standard practice. This gives a total inventory of 30 phonemes.

The copula root $*h_1$és- consists of three of these phonemes — $*h_1$, *e, *s — in a specific order. The probability calculation proceeds from this count.

3.2 Fortson (2004)

Fortson's reconstruction closely follows the laryngealist consensus and posits a consonantal inventory substantially identical to Beekes. The three laryngeals h_1, h_2, h_3 are retained as distinct phonemes. The vowel system likewise parallels Beekes. The total inventory is comparable to 30 phonemes, and $*h_1$és- remains a three-phoneme root under this reconstruction. The probability calculation yields the same result:

$$\mathbf{P(coincidental\ identity) = 1/30 \times 1/30 \times 1/30 = 1/27{,}000}$$

3.3 Meier-Brügger (2010)

Meier-Brügger similarly accepts the three-laryngeal system and produces an inventory of comparable size. Some organizational differences exist — notably in the treatment of the dorsal series and the status of certain resonants — but none of these affect the segments present in $*h_1$és-. The root remains h_1, e, s under this reconstruction, and the probability calculation is unchanged:

$$\mathbf{P(coincidental\ identity) = 1/30 \times 1/30 \times 1/30 = 1/27{,}000}$$

3.4 Szemerényi (1980)

Szemerényi's reconstruction represents the most conservative position among the major references. Working within a framework skeptical of laryngeals as independent phonemes, Szemerényi does not include h_1, h_2, h_3 as segments in his inventory. Under his reconstruction, the copula root appears as *és- — two phonemes rather than three.

This produces a different calculation. With laryngeals excluded, the inventory contracts to approximately 27 phonemes:

$$\mathbf{P(coincidental\ identity) = 1/27 \times 1/27 = 1/729}$$

At approximately 1 in 729, the probability of coincidental two-phoneme identity remains overwhelming as evidence against independent origin. The Szemerényi case therefore does not weaken the argument — it strengthens it. Even the most conservative reconstruction, which minimizes the phonemic inventory and reduces the root to two segments, still produces a probability of coincidental identity too low to be credible.

The argument is thus doubly robust. Under laryngealist reconstructions the probability of coincidental identity is approximately 1/27,000. Under the most conservative non-laryngealist reconstruction it is approximately 1/729. In neither case does the hypothesis of coincidental identity remain viable.

3.5 Summary Across Reconstructions

Reconstruction	Laryngeals	Total phonemes	Root	P(coincidental)
Beekes (1995)	Yes — 3	30	*$h_1$és- (3 phonemes)	1/27,000
Fortson (2004)	Yes — 3	30	*$h_1$és- (3 phonemes)	1/27,000
Meier-Brügger (2010)	Yes — 3	30	*$h_1$és- (3 phonemes)	1/27,000

Reconstruction	Laryngeals	Total phonemes	Root	P(coincidental)
Szemerényi (1980)	No	27	*és- (2 phonemes)	1/729

Note: The count of 30 phonemes for all laryngealist reconstructions reflects the standard consensus inventory of 25 consonants (including 3 laryngeals) plus 5 vowels (e, o, a, i, u), with long vowels treated as positional variants. This inventory is consistent across Beekes, Fortson, and Meier-Brügger, all of whom work within the laryngealist framework. The count of 27 for Szemerényi reflects the same inventory minus the 3 laryngeal phonemes.

3.6 Phonotactic Constraints

All four reconstructions agree on one further point relevant to the calculation: PIE root structure was subject to systematic phonotactic constraints that reduced the number of possible roots below the theoretical maximum. Iverson and Salmons (1992) and Corbeau (2013) document prohibitions against co-occurrence of plain voiced stops within a root, restrictions on voiceless and aspirated stop combinations, and a strong tendency against repeated identical consonants. These constraints apply regardless of which reconstruction is adopted, and they operate in the same direction in every case: by reducing the pool of available roots, they make the probability of any two roots accidentally matching lower than the raw calculation suggests.

The figures of 1/27,000 and 1/729 are therefore floors, not ceilings. The true probability of coincidental identity between the copula root and the pronoun stem is lower under every reconstruction considered here. The result of Section 4 is conservative by design.

Section 4. The Probability Calculation

The standard reconstruction of PIE assigns *$h_1$és- to two separate paradigmatic categories: the copula paradigm and the third-person pronoun paradigm. This paper argues that the probabilistic evidence does not support that separation. The question we ask is not whether two independently existing forms happen to be related, but whether the phonemic evidence is sufficient to justify treating them as two distinct morphemes in the first place.

This distinction in framing matters. The claim is not “these two similar-looking forms are probably historically connected” — a formulation that invites the response that connection can always be pushed back to a hypothetical earlier reconstructed stage. The claim is rather that the paradigmatic separation imposed by modern analysts on this material is probably artificial — that historical linguistics has been filing the same reconstructed element under two different cabinet labels, and that the probability argument reveals where that filing is unwarranted. What is being questioned is not PIE itself but the taxonomic practice that has organized its reconstruction.

To formalize the question: given a finite phonemic inventory of 30 segments, what is the probability that two independently reconstructed morphemes would be assigned identical three-phoneme sequences by coincidence? Under the simplest assumption — that each of the 30 phonemes in the PIE inventory is equally available in each root position — the calculation is:

$$\mathbf{P(coincidental\ identity) = 1/30 \times 1/30 \times 1/30 = 1/27{,}000}$$

This baseline figure of 1/27,000 requires refinement in light of two features of PIE phonology that the uniform distribution assumption ignores: the position-specific structure of PIE roots, and the observed frequencies of the relevant phonemes. Before developing those refinements, it is worth establishing the right framework for the calculation itself.

4.1 Probability and Statistics: A Necessary Distinction

The calculation in this section uses probability, not statistics. For readers more familiar with statistical methods, the distinction is worth making explicit, because the two frameworks ask different questions and are appropriate in different circumstances.

Statistics is the discipline of inference from samples. Given a body of observed data, statistical methods allow us to ask whether a pattern in that data is likely to reflect a real underlying regularity or merely the noise of random sampling. Statistical reasoning requires a sample drawn from a larger population, repeated observations, and assumptions about how variation is distributed. Its characteristic output is a p-value: the probability of observing results at least as extreme as those obtained, assuming the null hypothesis is true.

Probability, by contrast, is the discipline of reasoning about outcomes in a well-defined, finite universe. It does not require a sample or repeated observations. It requires only that the relevant universe be clearly specified — that we know what the possible outcomes are and can assign likelihoods to them. Its characteristic output is a simple fraction: the proportion of possible outcomes that constitute the event of interest.

PIE is not a statistical problem. There is no sample to be drawn, no population to generalize to, no experimental variation to account for. PIE is a reconstructed closed system with a finite, well-documented phonemic inventory. The question — how probable is it that two roots in this system share the same phonemic content by chance? — is a question about a defined universe of possible outcomes. That is a probability question, and probability is the right tool for it.

The classical theory of probability, developed by Pascal, Fermat, and later Laplace, was designed precisely for this kind of situation: a finite set of equally possible outcomes, from which we ask what proportion favor a particular event. Hacking (2001) provides a clear introduction to this framework and its distinction from frequentist statistics. The uniform distribution assumption used above — treating each of the 30 PIE phonemes as equally available in each root position — is the standard starting point for classical probability calculations of this kind. It is not a statistical approximation of some underlying empirical distribution; it is a principled baseline for a combinatorial argument about a closed system.

This matters for how the refinements below should be understood. When we adjust for observed phoneme frequencies and root structure, we are not running a statistical test — we are refining the probability model by replacing the uniform assumption with better-grounded estimates of the relevant likelihoods. The result is still a probability calculation, not a statistical inference, and it should be evaluated accordingly: not by asking whether it meets some conventional significance threshold, but by asking whether the probability of coincidental identity is low enough to make coincidence an implausible explanation.

4.2 Refining the Calculation

Two features of PIE root phonology are relevant to refining the baseline calculation.

Root structure. PIE roots are overwhelmingly CVC in structure — a consonant, a vowel, a consonant — and the vowel position is far from equally distributed across the five PIE vowels. The vowel *e is the default grade vowel of the ablaut system and appears in the full grade of the majority of PIE roots (Beekes 1995; Fortson 2004). Treating *e as equally probable with *o, *a, *i, and *u, as the uniform assumption does, understates its actual frequency. Conversely, the initial position is restricted to consonants only — not all 30 phonemes — and the final position is similarly consonant-dominated. The uniform assumption treats all three positions as if they draw from the same 30-phoneme pool, which does not reflect the actual structure of PIE roots.

Phoneme frequency. Within each position, phonemes are not equally frequent. Hartmann (2021), drawing on trigram data from Wiktionary PIE reconstructions (n = 6,236 trigrams), shows that $*h_1$ appears as a root-initial phoneme in only approximately 3.2% of attested trigrams — making it a marked and infrequent root-initial segment. For the final position, Iverson and Salmons (1992) show that only 14% of reconstructed roots are stop-final, leaving the majority to end in resonants, laryngeals, or the single PIE fricative *s, which appears in approximately 10–15% of root-final positions.

The frequency-adjusted calculation. Substituting observed frequencies for the uniform assumption gives a more realistic estimate. Using $*h_1$ as root-initial in approximately 3.2% of cases (Hartmann 2021), *e as the default medial vowel in approximately 60% of full-grade roots — a conservative estimate given its status as the dominant ablaut vowel — and *s in final position in approximately 12% of roots, the calculation is:

$$\mathbf{P(frequency\text{-}adjusted) = 0.032 \times 0.60 \times 0.12 \approx 1/13{,}000}$$

At approximately 1 in 13,000, the frequency-adjusted estimate falls between the uniform baseline of 1/27,000 and lower estimates that would result from treating *e as fully predetermined. It is the more defensible single figure: it acknowledges that *e is the dominant but not the only PIE root vowel, uses empirically grounded frequency data for the initial and final consonants, and avoids both overstating and understating the improbability of the match.

The uniform baseline of 1/27,000 should therefore be understood as an upper bound on the probability of coincidental identity — the figure produced under the most favorable assumptions for the opposing hypothesis. The frequency-adjusted figure of approximately 1/13,000 is the more realistic estimate. Both figures are well below any threshold at which coincidence would be a credible explanation.

4.3 Sensitivity Analysis

The following table summarizes the probability estimates under varying assumptions, to give the reader a clear picture of how robust the conclusion is across different choices of input.

Scenario	Key Assumptions	P(coincidental)	Odds (1 in X)
Uniform (baseline)	30 phonemes, equal probability	1/27,000	27,000
Frequency-adjusted (preferred estimate)	*h_1 initial ≈3.2%, *e medial ≈60%, *s final ≈12%	≈1/13,000	~13,000
Conservative (lower *e frequency)	*h_1 initial ≈3.2%, *e medial ≈50%, *s final ≈10%	≈1/6,200	~6,200
Phonotactics only (smaller root pool)	~5,000–8,000 effective legal roots	1/5,000–1/8,000	5,000–8,000

*Note: Frequency estimates draw on Hartmann (2021) and standard PIE ablaut descriptions. The phonotactics-only row reflects a smaller effective root pool due to co-occurrence constraints (Iverson and Salmons 1992; Corbeau 2013), which raises the collision probability but does not account for the marked nature of the specific sequence *$h_1$és-. All estimates support the same conclusion: coincidental identity between the copula root and the pronoun root is not a credible explanation.*

4.4 The Scope of the Objection

A natural objection presents itself: does not the same calculation apply to any two identical forms in PIE? Would not any pair of matching roots yield a similar figure, making the test trivially applicable to any accidental homophony?

The objection misidentifies what the test establishes. The probability calculation does not, by itself, prove that the paradigmatic distinction is artificial. It establishes that coincidental identity is sufficiently improbable to require explanation — and that the explanation on offer from the received framework, namely that two independent morphemes simply happen to be phonemically identical, is not available at this probability level. What motivates the conclusion that the distinction is taxonomic rather than linguistic is the convergence of three independent lines of evidence, no one of which alone would be decisive:

> *(i) The phonemic identity itself, with a probability of coincidental occurrence of approximately 1/13,000 under the preferred frequency-adjusted estimate, or 1/27,000 under the uniform baseline.*
>
> *(ii) The functional relationship between the two categories. The copula and the pronoun are not arbitrary vocabulary items that happen to sound alike. Both belong to the domain of reference and predication rather than description: the pronoun points at an entity; the copula asserts its identity or existence. The semantic and functional alignment between these categories is independently motivated (Katz 1996) and constitutes a prior expectation that they might reflect a single underlying element that grammaticalized into both functions, or perhaps never fully differentiated between them.*
>
> *(iii) The persistence of the identity across all attested daughter languages without exception — a pattern that has no explanation under the hypothesis of coincidence, since coincidence does not replicate itself systematically across independent branches. Under the hypothesis that the distinction is taxonomic rather than linguistic, the persistence is exactly what we would expect: analysts in each branch inherited and perpetuated the same filing system, while the forms themselves remained what they had always been.*

The test is designed for precisely this kind of convergent case. Where all three conditions hold simultaneously, the conclusion that the paradigmatic separation is an artifact of analytical practice is not merely probable — it is, on any reasonable standard of inference, the only explanation available.

Section 5. The Copula and Pronoun Paradigms

The claim that the paradigmatic separation of *$h_1$és- into a copula paradigm and a pronoun paradigm is taxonomic rather than linguistic rests on evidence from six IE daughter languages: Sanskrit, Greek, Avestan, Latin, Hittite, and Old Irish. Each branch is examined in turn. A comparative summary and cross-linguistic parallel follow.

The primary exhibit in each language is the third-person paradigm, where the *$h_1$és- root is most transparently visible. The root surfaces differently depending on phonological environment: where a case or person ending follows the root, the final *-s is preserved (*$h_1$es-); in the bare nominative, where nothing follows the root, final *-s was regularly lost (*h_1e-). This is an ordinary phonological alternation, not evidence of two different roots. The argument does not require *$h_1$és- to surface unreduced throughout the entire pronoun system — only that the element doing third-person pronominal work and the element doing copular work are the same root, and that this identity persists across independent branches.

5.1 The PIE Oblique Pronoun Paradigm

The most direct evidence does not come from a daughter language at all, but from the reconstructed PIE third-person pronoun paradigm itself. The nominative stem of the third-person pronoun is reconstructed as *h_1e- (or *h_1i- in some forms), with no case ending following the root in absolute final position — the environment in which final *-s is regularly lost. In the oblique cases, where a case ending follows the root, the full sequence

*$h_1$es- is preserved, identical to the copula root. The nominative and oblique forms are therefore not two different roots, but the same root *$h_1$és- surfacing differently according to a regular and independently motivated phonological environment.

Case	Pronoun (oblique stem)	Copula
Nominative	***h_1e**- (*-s lost in final position)	—
Genitive	***$h_1$és**o	—
Ablative	***$h_1$es**mōd	—
Dative	***$h_1$es**mōi	***$h_1$es**mi (1sg)
Locative	***$h_1$es**mi	***$h_1$es**ti (3sg)

In the genitive, ablative, dative, and locative, the pronoun stem is *$h_1$es-, the same three phonemes in the same order as the copula root *$h_1$és-. The nominative *h_1e- is the same root with its final *-s lost by regular sound change in absolute final position — the position the bare nominative occupies, with no ending following. The dative and locative forms *$h_1$esmōi and *$h_1$esmi are particularly striking: they differ from the copula's *$h_1$esmi (1sg present) and *$h_1$esti (3sg present) only in their endings, not in the root itself.

This is the tightest parallel available anywhere in the data, because it requires no appeal to daughter-language reflexes or sound changes across branches. It is a relationship internal to the PIE reconstruction itself: the same three-phoneme root, *$h_1$és-, appears in both the copula paradigm and the pronoun paradigm, surfacing as *$h_1$es- wherever an ending follows and as *h_1e- in the bare nominative, where regular final-*s loss applies. The daughter language evidence surveyed in the remainder of this section shows that this identity persists — in various forms — across the attested branches.

5.2 Sanskrit

Sanskrit has no dedicated third-person personal pronoun; demonstratives serve that function. The proximal demonstrative idam is the primary exhibit. Its oblique stem as- reflects *$h_1$es- directly, with all three phonemes of the root intact; the nominative ayam/iyam/idam reflects the same root with regular loss of final *-s.

*Proximal demonstrative idam (oblique stem as- = *$h_1$és-)*

*Note: Sanskrit as- represents all three phonemes of *$h_1$és-. The laryngeal *h_1 and the vowel *e merge to Sanskrit a; *s is preserved before the case endings. The bolded as- in the table below is therefore the full three-phoneme root, not a two-phoneme subset of it.*

Case	Masc	Fem	Neut
Nom sg	ayam	iyam	idam
Gen sg	**as**ya	**as**yāḥ	**as**ya
Dat sg	**as**mai	**as**yāi	**as**mai
Loc sg	**as**min	**as**yām	**as**min

Copula as-

Person	Present
1sg	**as**mi
2sg	**as**i
3sg	**as**ti
1pl	smaḥ
2pl	stha
3pl	santi

*3sg asti directly reflects PIE *$h_1$ésti. The genitive asya and dative asmai of the demonstrative sit alongside asti with the full *$h_1$és- root visible in both.*

5.3 Greek

Greek presents a partial exhibit. The copula reflects *$h_1$és- throughout, though intervocalic *s was lost by a Greek-internal sound law in 1sg eimi and 3pl eisi (from *$h_1$ésmi, *$h_1$senti), while *s is preserved before *t in esti, esmen, este. The demonstrative system has undergone t- innovation, and the root *$h_1$és- is not prominent in the surviving demonstratives. Greek is one witness among six, and the copula alone is sufficient to establish the root's continuity in this branch.

Copula εἰμί (eimi) — all forms from *$h_1$és-

Person	Present	Greek	From
1sg	**ei**mi	εἰμί	*$h_1$ésmi (*s lost)
2sg	**ei**	εἶ	*$h_1$ési (*s lost)
3sg	**es**ti	ἐστί	*$h_1$ésti
1pl	**es**men	ἐσμέν	*$h_1$ésmen
2pl	**es**te	ἐστέ	*$h_1$éste
3pl	**ei**si	εἰσί	*$h_1$ésenti (*s lost)

*Every form in this paradigm derives from *$h_1$és-. Where *s appears between vowels (1sg, 3pl) it was lost by a regular Greek sound law; where it stood before a consonant (3sg, 1pl, 2pl) it was preserved. 3sg esti directly reflects PIE *$h_1$ésti with the full root intact.*

5.4 Avestan

Avestan is among the clearest exhibits. The copula ahmi (1sg) and the genitive pronoun ahya both reflect the full root *$h_1$és-: intervocalic *s regularly became h in Avestan, so *$h_1$esmi and *$h_1$esya surface as ahmi and ahya. All three phonemes of the root are present, with *s realized as h.

Demonstrative ha/hā/tat — primary 3rd person

Case	Masc	Fem	Neut
Nom sg	hō	hā	tat
Gen sg	**ah**ya	**aŋ**hiyā	**ah**ya

Case	Masc	Fem	Neut
Dat sg	**ah**māi	**aŋ**hāi	**ah**māi
Nom pl	tōi	tāh	tā

Copula

Person	Present
1sg	**ah**mi
2sg	**ah**i
3sg	**as**ti
1pl	m**ah**i
3pl	hənti

*ahmi (I am) and ahya (of him/that) both reflect *$h_1$ésmi and *$h_1$ésya, the full three-phoneme root with *s > h, doing copular and pronominal work simultaneously. 3sg asti preserves *s unchanged before *t. This is among the most direct attestations of root identity in the daughter language record.*

5.5 Latin

Latin presents a two-part exhibit with an important asymmetry. The third-person pronoun is/ea/id is built throughout on the stem *h_1e-/*h_1i-, without the final *-s of the copula root: forms such as eius, eō, and the oblique cases generally show no trace of *s. The copula esse, by contrast, preserves the full root *$h_1$és- in es/est/estis; sum/sumus/sunt show regular loss of *s before m, phonologically expected and not a problem for the argument. Latin therefore illustrates the two-phoneme element *h_1e- in the pronoun and the full three-phoneme root *$h_1$és- in the copula side by side — the same relationship as the PIE nominative/oblique alternation in Section 5.1, with the pronoun here generalizing the reduced *h_1e- stem across its paradigm rather than restricting it to the nominative.

3rd person pronoun is/ea/id (stem *h_1e-/*h_1i-)

Case	Masc	Fem	Neut
Nom sg	**i**s	**e**a	**i**d
Acc sg	**e**um	**e**am	**i**d
Gen sg	**e**ius	**e**ius	**e**ius
Dat sg	**e**ī	**e**ī	**e**ī
Abl sg	**e**ō	**e**ā	**e**ō
Nom pl	**e**ī/iī	**e**ae	**e**a
Acc pl	**e**ōs	**e**ās	**e**a
Gen pl	**e**ōrum	**e**ārum	**e**ōrum

Copula esse (root *$h_1$és-)

Person	Present
1sg	sum
2sg	**es**
3sg	**es**t
1pl	sumus
2pl	**es**tis
3pl	sunt

*is, ea, id, eius, eō throughout show the two-phoneme element *h_1e- (or *h_1i-) without *s. est directly reflects the full root *$h_1$ésti. es/est/estis preserve *$h_1$es- intact.*

5.6 Hittite

Hittite is the anchor exhibit. As the earliest attested IE language, it preserves laryngeals as actual consonants in the cuneiform record. The copula root ēš- directly attests *$h_1$és- with the laryngeal's presence readable off the text. No reconstruction is required.

Copula eš-/aš- (THE ANCHOR EXHIBIT, root *$h_1$és-)

Person	Present	Preterite
1sg	**ēš**mi	**ēš**un
2sg	**ēš**ši	**ēš**ta
3sg	**ēš**zi	**ēš**ta
1pl	**ēš**weni	**ēš**wen
2pl	**ēš**teni	**ēš**ten
3pl	**aš**anzi	**ēš**er

ēš- reflects *$h_1$és- with the laryngeal's vocalic effect visible in the long ē. All three phonemes of the root — h_1 (as vowel length), e, and š — are present. 3pl ašanzi shows short a- reflecting the same root in an unstressed syllable, where the vowel was reduced — phonologically expected and itself evidence of the root's behavior under different stress conditions.

5.7 Old Irish

Old Irish has grammaticalized the equational/existential distinction into two separate verbs, making the functional claim about *$h_1$és- directly visible. The copula is handles identificational predication; at·tá handles locational and existential predication. The *$h_1$és- reflex was assigned to the equational slot.

3rd person pronoun (stem *h_1e-, *-s not present)

	Masc	Fem	Neut
Nom sg	**é**	sí	**ed**

Copula is — equational/identificational (root *$h_1$és-)

Person	Present	Preterite	From
1sg	**am**	ba	*$h_1$ésmi (*s lost before m)
2sg	**at**	bat	*$h_1$ési
3sg	**is**	ba/bu	*$h_1$ésti
1pl	**am**mi	bammi	*$h_1$ésmi

Person	Present	Preterite	From
2pl	adib	babir	—
3pl	**it**	batar	*$h_1$énti

Substantive verb at·tá — existential/locational

Person	Present
1sg	táu
2sg	taí
3sg	at·tá
1pl	táam
2pl	taaid
3pl	táat

*3sg is directly reflects PIE *$h_1$ésti with the full root intact. 1sg am and 1pl ammi reflect *$h_1$ésmi with regular Celtic loss of *s before m; 2sg at reflects *$h_1$ési. The pronoun é (masc 3sg) and ed (neut) show only *h_1e-, without *s — the same reduced stem found in the Latin pronoun and in the PIE nominative. at·tá derives from ad- + *steh$_2$- 'to stand' — the same root as Spanish estar. The parallel is structural, not inherited.*

5.8 Comparative Summary

The *$h_1$és- root across copula and pronominal systems in six IE branches.

Language	3sg Copula (*$h_1$és-)	Pronominal forms	Notes
Sanskrit	asti	asya, asmai (*$h_1$és-, oblique)	Proximal demonstrative oblique stem and copula share the full root
Greek	esti	— (t- dominant)	Copula preserves *$h_1$és- throughout; demonstrative innovated
Avestan	asti	ahya, ahmāi (*$h_1$és-, *s>h)	Clearest exhibit: ahmi alongside ahya, both from *$h_1$és-
Latin	est	is, ea, id, eius (*h_1e-, *s absent)	Copula has full root; pronoun generalizes reduced *h_1e- stem

Language	3sg Copula (*h$_1$és-)	Pronominal forms	Notes
Hittite	ēšzi	— (apā- secondary)	Anchor: full root *h$_1$és- directly attested in ēš-
Old Irish	is	é, ed (*h$_1$e-, *s absent)	Copula has full root; two-be system assigns *h$_1$és- to equational slot

Every copula in this table reflects the full root *h$_1$és-. In the pronoun paradigms, Sanskrit and Avestan show the same full root *h$_1$és- in the oblique stem, while Latin and Old Irish generalize the reduced stem *h$_1$e- (the same reduction found in the PIE nominative, where final *-s is regularly lost) across their pronoun paradigms. No branch shows the copula root and the third-person pronominal root as phonemically distinct sources — in every case, the pronoun reflects either the full root *h$_1$és- or its regular reduction *h$_1$e-. The identity is a structural feature of the reconstruction, not a local coincidence.

5.9 The Two-Be Typology

Where languages have independently grammaticalized a distinction between two verbs of being, the *h$_1$és- reflex is consistently assigned to the equational and identificational slot.

Language	From *h$_1$és-	From *steh$_2$-	Functional division
Old Irish	is (equational)	at·tá (existential)	Grammaticalized in Proto-Celtic
Spanish	ser (essential)	estar (locational)	Independent Ibero-Romance innovation
Portuguese	ser	estar	Parallel to Spanish; independent

Old Irish and Spanish arrived at this division independently. Both assigned *h$_1$és- to the equational slot and recruited *steh$_2$- for the existential slot. This convergence suggests a stable cognitive association between this root and pronominal/equational predication across time and geography.

5.10 The Hebrew Cross-Linguistic Parallel

The cognitive alignment of pronominal material with copular function is not confined to the IE family. Katz (1996) establishes an independent pronoun-copula root identity in Biblical Hebrew, set out below. The PIE column gives $^*h_1e$-, the reduced form of the root found in the PIE nominative and in several daughter-language pronoun paradigms (Section 5.8), since it is this reduced form that corresponds typologically to the Hebrew pronominal material; the point of the comparison is the functional alignment of pronoun and copula, not a claim of phonemic identity between PIE and Hebrew.

Function	PIE	Hebrew/Semitic
Copula	$^*h_1$és-	hāwāh / hayāh
3rd person pronoun	$^*h_1e$-	hū (masc) / hī (fem)
Distal demonstrative (pronoun + def. art.)	$^*h_1e$-	hahū (masc) / hahī (fem)

PIE and Proto-Semitic are unrelated language families. The pronoun-copula root identity in Hebrew cannot be a shared inheritance from PIE. Its independent existence suggests that the cognitive alignment of pronominal reference with predicational being is a recurring feature of human linguistic organization, not a PIE-specific accident.

Section 6. Evidence from Daughter Languages: Addressing the Turtles-All-the-Way-Down Objection

6.1 The Objection

The most durable response to any claim of root identity within PIE is to push the claim back to an earlier hypothetical stage of the language. If $^*h_1$és- appears to do double duty as both copula and pronoun in the reconstruction, a critic can propose that at some still earlier stage — prior to the reconstructed proto-language itself — the two functions were carried by distinct, coincidentally similar forms, which only later collapsed into apparent identity. Since no stage earlier than PIE is directly recoverable, this move can be repeated indefinitely: any claimed identity can always be attributed to an unrecoverable prior merger rather than a genuine original unity. This is the turtles-all-the-way-down problem named in this paper's working notes.

6.2 Why Persistence Across Branches Defeats the Objection

The objection has force only if the identity in question is fragile — confined to the reconstructed proto-form and dependent on a single analytical choice. It loses its force if the identity instead turns out to be robust: if it persists, independently, across multiple branches that separated from each other thousands of years ago and underwent entirely different sound changes thereafter.

This is precisely what Section 5 demonstrated. The *$h_1$és- identity is not an artifact visible only in the reconstructed proto-form. It surfaces in Sanskrit, where the oblique demonstrative stem as- and the copula as- are visibly the same material. It surfaces in Avestan, where ahmi and ahya sit side by side in the synchronic grammar. It surfaces in Hittite, the most archaic attested branch, where the laryngeal is read directly off the cuneiform record rather than reconstructed. It surfaces, in reduced form, in Latin and Old Irish, where the pronoun generalizes the same *h_1e-reduction independently found in the PIE nominative.

A critic invoking the turtles-all-the-way-down move must now explain not merely why the PIE reconstruction shows this identity, but why every daughter branch, independently, continued to show it — despite millennia of separate development, contact, and sound change. The hypothesis of an early coincidental merger does not predict this. If the merger were coincidental and external to the actual grammar of PIE, there would be no mechanism compelling every daughter language to preserve and continue exhibiting the same coincidence after the languages had gone their separate ways. Coincidences, by their nature, do not replicate themselves systematically across independent lines of transmission.

6.3 The Spanish Illustration

A vivid, non-technical illustration of the same persistence is available outside the directly inherited evidence, in the Ibero-Romance development of the two-be system. Spanish ser and estar did not exist as a binary distinction in Latin; the distinction is a medieval Ibero-Romance innovation. Yet when Spanish independently developed two verbs of being, it assigned the reflex of *$h_1$és- (via Latin esse) to the equational and identificational slot, exactly paralleling the division Old Irish had already made, independently, over a thousand years earlier and a thousand miles away.

The phrase eso es ("that is") places the demonstrative and the equational copula side by side, both visibly built on the same etymological material, in a language separated from PIE by roughly six thousand years of transmission and any number of intervening mergers, splits, and innovations. If the identity between pronominal/demonstrative reference and the copula were merely a coincidental PIE-internal artifact, there is no reason it should resurface, independently, in a grammaticalization event that took place in medieval Iberia.

6.4 Why Pushing the Coincidence Back Does Not Resolve It

The deeper response to the turtles-all-the-way-down objection is this: pushing the identity back to an earlier hypothetical stage does not eliminate the improbability calculated in Section 4 — it relocates it. If a critic proposes that the identity arose from an earlier coincidental merger of two originally distinct forms, that merger itself requires an explanation.

A coincidental phonological merger between a copula root and a pronominal root, occurring at whatever stage one wishes to posit, is exactly the kind of event whose improbability the calculation in Section 4 addresses. Renaming the stage at which the coincidence is supposed to have occurred does not make the coincidence more probable; it only removes it one step further from direct evidence, where it becomes correspondingly harder to defend rather than easier.

6.5 Summary

The daughter language evidence therefore does not merely supplement the PIE-internal argument of Section 5.1 — it closes off the most obvious avenue of escape from it. The identity is not confined to one reconstruction, one branch, or one analytical tradition. It persists across six independently developing branches, including the most archaic directly attested branch, and it resurfaces independently in a medieval grammaticalization event unconnected to the original PIE pattern by anything other than common descent. An objection that requires every one of these independent developments to be separately coincidental is not a parsimonious alternative to the hypothesis of common origin; it is a considerably less probable one.

Section 7. Reconstruction Independence

Section 3 demonstrated that the probability calculation yields essentially the same conclusion under each of four major reconstructions of PIE: Beekes (1995), Fortson (2004), Meier-Brügger (2010), and Szemerényi (1980). This section makes explicit what that result establishes methodologically, and why it matters for the argument as a whole.

7.1 The Standard Objection Restated

The most readily available response to any claim about PIE root identity is to push the claim back to an earlier hypothetical stage of the reconstruction. If the copula and the third-person pronoun appear identical under one reconstruction, a critic can propose an alternative reconstruction under which they are not — and the burden then falls on the proponent to defend a particular reconstruction against its competitors, a debate that can be deferred indefinitely.

This objection has force only if the result is sensitive to which reconstruction is adopted. If the $*h_1$és- identity disappears under some major reconstructions but not others, then the argument reduces to a dispute about which reconstruction is correct — a dispute this paper is not equipped to settle, and need not settle.

7.2 The Result Across Reconstructions

Section 3 showed that this sensitivity does not arise. Under Beekes, Fortson, and Meier-Brügger — the three major reconstructions that retain the laryngeal series — the copula root and the pronoun stem both consist of the same three phonemes, h_1, e, s, in the same order, drawn from an inventory of 30 phonemes. The probability of this identity arising by chance is approximately 1/27,000 under each of these three reconstructions independently.

Szemerényi's reconstruction, which does not treat the laryngeals as independent phonemes, might appear to offer the escape route the objection requires: without h_1, the identity reduces from three phonemes to two. But as Section 3 showed, this reduction does not eliminate the identity — it merely reduces the inventory to 27 phonemes and the shared material to *és-, yielding a probability of coincidental identity of approximately 1/729. The identity persists; only its magnitude changes.

7.3 What Reconstruction Independence Establishes

The result is therefore not contingent on resolving the laryngealist debate, nor on adjudicating between the specific phonological analyses of Beekes, Fortson, Meier-Brügger, or Szemerényi. Whichever of these reconstructions a reader adopts — including the one most likely to be invoked as a way of dissolving the identity — the *$h_1$és- identity remains, and remains improbable as a chance occurrence.

This converts the standard objection into something the present argument does not need to engage. A critic who wishes to dispute the conclusion cannot do so simply by preferring a different reconstruction, because the conclusion has already been shown to hold under the reconstruction such a critic would be most likely to prefer. The argument does not depend on a choice between competing reconstructions; it survives the full range of choices currently available in the field.

7.4 Scope of the Claim

This independence is necessarily bounded by the reconstructions actually surveyed. The claim is not that no conceivable future reconstruction could eliminate the $h_1$és- identity, but that no reconstruction currently representative of the field does so. Should a fundamentally different phonological architecture for PIE emerge — one that, for instance, analyzed the copula root and the pronoun stem as drawing on entirely distinct segments not captured by any of the four reconstructions considered here — the present argument would need to be reassessed in light of it. No such reconstruction is currently part of the scholarly mainstream, and the four reconstructions surveyed here represent the range of serious alternatives a reviewer would be likely to raise.

Section 8. Engagement with Existing Literature

8.1 The Standard Account of the PIE Copula

The dominant position in the literature treats the PIE verb “to be” as a suppletive paradigm built from multiple distinct roots. The standard account identifies four contributing stems across the IE family: *$h_1$es- (the present-tense copula examined throughout this paper), *steh$_2$- (“to stand,” source of Spanish estar, Old Irish at·tá), *wes- (source of English was, were), and *b^{h}uH- (“to become,” source of English be, Latin fuī). Within this suppletive system, *$h_1$es- is consistently treated as the core present-tense element, restricted to the present and closely related forms, with other stems filling in the past tense and other gaps left by its limited paradigm (Irslinger 2018).

This paper does not dispute the suppletive architecture of the IE copula system as a whole. Its claim is narrower: it concerns only the etymological source of *$h_1$es- itself, not the broader suppletive paradigm into which *$h_1$es- was integrated.

8.2 The Puzzle of *$h_1$es-'s Isolation

A feature of *$h_1$es- noted in the literature, though not extensively explored, is its isolation within the PIE lexicon. Unlike the great majority of PIE roots, which are independently attested as full lexical verbs with their own semantic content elsewhere in the language, *$h_1$es- has no other attested lexical use. Its only reconstructable meaning, in every branch, is “to be” or “to exist.” Irslinger (2018) notes that this may not reflect its original meaning, though no consensus account of an earlier, independent lexical source for *$h_1$es- has been established in the literature surveyed here.

This isolation is significant for the present argument. A root with no independent lexical history, restricted entirely to copular function, is precisely what one would expect if the root’s origin lay not in ordinary lexical derivation but in the repurposing of pronominal material — material that, by its nature, does not behave like an ordinary noun or verb root and would not be expected to surface independently as one.

8.3 The Standard Account of the PIE Pronoun System

The literature on PIE pronouns recognizes *h_1e- as one of several pronominal and demonstrative stems reconstructed for the proto-language, existing alongside *so-/*to- (the source of the distal/anaphoric demonstrative that became dominant in Greek, among others), *ḳe- (a proximal deictic), and other minor stems. Beekes (1995) reconstructs *h_1e- specifically as an anaphoric pronoun, glossed as referring to an entity "just named" in discourse — distinct from, though coexisting with, the more general demonstrative system built on *so-/*to-.

This paper's argument concerns specifically the *h_1e- paradigm, not the *so-/*to- system. The two pronominal stems are not in competition as candidates for identity with the copula: only *h_1e- shares its phonemic material with *$h_1$és-, and the existing literature's recognition of *h_1e- as an independently established anaphoric pronoun, separate from the demonstrative system proper, is consistent with treating it as a discrete paradigmatic unit whose identity with the copula root can be assessed on its own terms.

8.4 A Prior Proposal and Its Rejection

A pronominal origin for *$h_1$es- has in fact been raised before, though not developed into the kind of argument advanced in this paper. Irslinger (2018) notes that a derivation of *$h_1$es- from pronominal material is attractive on functional grounds, since it would explain why *$h_1$es- is restricted to the present tense and lacks any independent lexical use elsewhere — precisely the isolation discussed in Section 8.2 above. Irslinger reports, however, that this etymology is considered problematic on formal grounds: citing Dunkel (2014), she notes that more recent reconstructions analyze the relevant demonstrative material as the stem *e-sm-, which is not treated as homonymous with the verb *$h_1$es-.

This is a genuine prior engagement with the idea explored in this paper, and it must be addressed directly. The objection, as reported, rests on a specific claim: that the demonstrative material relevant to comparison is best segmented as *e-sm-, a unit in which the *s belongs with the following formant rather than with the preceding *h_1e-, and which is accordingly treated as a different phonological object from the verb root *$h_1$es-, in which the *s is root-final.

8.5 Why the Formal Objection Does Not Resolve the Question

The *e-sm- segmentation is a morphological analysis, not a phonemic one. It reflects a decision about where, within the attested oblique pronoun forms (*$h_1$esmōi, *$h_1$esmi, and so on), the boundary between root and formant should be drawn. But the probabilistic argument developed in this paper does not depend on where that boundary is drawn. The calculation in Section 4 concerns the sequence of phonemes actually present — h_1, e, s — not the morphological labels subsequently assigned to portions of that sequence. Whether the *s in *$h_1$esmōi is analyzed as root-final or as the initial consonant of a following formant *-sm-, the phonemic sequence h_1-e-s is present in both the pronoun and the copula in exactly the same order. The segmentation question is a question about historical morphological structure; the probability question is a question about phonemic identity. The two are independent, and a morphological re-segmentation does not by itself change the probability that the phonemic identity is coincidental.

Put differently: even granting that *-sm- might be analyzed as a unitary oblique formant rather than as root *s plus case-marking *m, this formant itself begins with the same *s that ends the copula root, immediately following the same *h_1e- that begins it. The objection relocates the *s from one side of a morpheme boundary to the other; it does not remove the *s from the environment immediately following *h_1e-, nor does it explain why a formant beginning in *s would have attached, across multiple oblique cases, specifically to the same pronominal stem that is also the one apparently sharing a root with the copula. The formal objection identifies a genuine question about morphological segmentation; it does not, on inspection, supply an alternative account of why the phonemic sequence h_1-e-s recurs in both paradigms at a rate the probabilistic method shows to be inconsistent with chance.

This paper's contribution, relative to this prior exchange, is therefore twofold. First, it shows that the question can be settled independently of the morphological segmentation dispute, by a probabilistic method that takes the phonemic sequence itself, rather than its morphological analysis, as the object of comparison. Second, it supplies the cross-branch and PIE-internal evidence developed in Sections 5 and 6, which was not part of the exchange reported by Irslinger and which independently supports the conclusion that the formal objection alone does not settle.

8.6 Relation to General Theories of Copula Origins

More broadly, typological and grammaticalization literature on copulas across the world's languages has long recognized that copulas frequently grammaticalize from sources outside ordinary lexical verbs — from demonstratives, from pronouns used in cleft-like constructions, and from other grammatical material whose original function was deictic or referential rather than predicative. The general typological plausibility of a pronoun-to-copula grammaticalization pathway is therefore well established as a cross-linguistic possibility. As Section 8.4 shows, the specific application of this pathway to *$h_1$es- has been raised before and met with a formal objection; this paper's contribution is to supply the probabilistic and cross-branch evidence needed to evaluate that proposal on grounds independent of the morphological segmentation dispute on which the prior exchange turned.

Section 9. Implications

9.1 For Synchronic Internal Reconstruction

The central methodological claim of this paper is that probabilistic analysis can resolve relatedness questions in closed systems where the standard comparative method, requiring written historical attestation, is unavailable. PIE offers a clear test case: no earlier written stage exists, and the turtles-all-the-way-down objection has traditionally made internal claims of this kind difficult to defend on comparative grounds alone. The method developed here does not replace the comparative method; it supplements it for exactly the cases the comparative method cannot reach — relationships internal to a single reconstructed system, prior to any attested divergence.

This suggests a more general methodological point. Wherever a proto-language's phonemic inventory and core vocabulary can be reliably established, the same probabilistic approach could in principle be applied to other suspected cases of internal root identity that have been set aside as undecidable. The contribution is not limited to the *$h_1$és- case; it is a method that other internal reconstruction questions, in PIE or in other proto-languages, could adopt.

9.2 The Cognitive Link Between Pronoun and Copula

The functional motivation for treating the copula and the third-person pronoun as historically connected was developed independently of the probabilistic argument, in Katz (1996), which argues on typological and grammaticalization grounds for a cyclical relationship between pronominal reference and copular predication across languages. The copula does not describe an action or a property; it asserts identity or existence, which is fundamentally what a pronoun does — point at something rather than characterize it. This functional affinity, established independently of any

PIE-specific evidence, is part of what makes the probabilistic finding in this paper plausible rather than merely surprising: the copula and pronoun are not arbitrary categories that happen to share phonemic material, but two categories with an independently motivated semantic kinship.

9.3 The Hebrew Parallel as Cross-Linguistic Confirmation

Section 5.10 showed that Biblical Hebrew exhibits an independent pronoun-copula root identity, established by Katz (1996) prior to and independently of the PIE argument developed here. PIE and Proto-Semitic are unrelated families; whatever produced the Hebrew pattern cannot be a shared inheritance from PIE, and whatever produced the PIE pattern cannot be a borrowing from Semitic. The two cases therefore constitute independent evidence for the same underlying tendency: that the cognitive association between pointing at an entity and asserting its existence or identity is not a PIE-specific accident but a recurring feature of how languages organize this part of the grammar.

9.4 The Two-Be Typology

Section 5.9 showed that languages independently developing a two-be system consistently assign the *$h_1$és- reflex, or its descendant, to the equational and identificational slot rather than the existential or locational one. Old Irish and Spanish made this assignment independently, separated by roughly a thousand years and a thousand miles, with no possibility of direct influence on each other's grammaticalization. This convergence supports the same conclusion as the Hebrew parallel from a different angle: not cross-family comparison, but repeated independent innovation within a single family, consistently landing on the same functional assignment. Both lines of evidence point toward a stable, non-accidental association between pronominal/identificational material and the equational copula.

9.5 Would PIE Speakers Have Perceived This Identity?

A further implication concerns whether PIE speakers themselves would have perceived *$h_1$és- as a single root doing double duty, or whether the identity would have been opaque to them in the way it has been to modern reconstruction.

Katz (2001) demonstrates that speakers' ability to perceive root identity across morphologically related forms depends not on the transparency of any individual item, but on the overall systematicity of derivation in the language as a whole. Comparing Hebrew, Mandarin, and English, the study shows that Hebrew and Mandarin speakers readily perceive root relationships even across discontinuous morphemes or phonologically independent syllables, because derivation in both languages is highly systematic, despite their typological differences. English speakers, by contrast, frequently fail to perceive the relationship between transparent forms such as roost and rooster, because English derivational morphology as a whole lacks comparable systematicity. The transparency of any single item, in other words, is less important than whether the language as a whole primes its speakers to expect root identity.

PIE, on the standard reconstruction, was a highly systematic root-based language, with derivational and inflectional morphology built transparently from a limited inventory of roots through productive ablaut and affixation. If the typological generalization established in Katz (2001) holds, PIE speakers would have been in a position closer to that of Hebrew or Mandarin speakers than to that of English speakers: root identity across related forms would have been psychologically salient to them, not opaque. The paradigmatic separation of *$h_1$és- into distinct copula and pronoun "morphemes," in other words, may be an opacity introduced by the comparative method itself — a product of reconstruction across millennia of daughter-language divergence — rather than a feature that would have been opaque to speakers of PIE at any stage of its existence.

This reframes the central claim of Section 4. The taxonomic artifact identified there is not only a feature of modern scholarly categorization; it may be a feature that modern categorization has retrojected onto a system whose own speakers, by the typological evidence available, would most likely not have recognized it as a separation at all.

9.6 Broader Questions of Root Identity in PIE

If the case developed in this paper is accepted, a further question follows: how many other pairs or sets of forms currently treated as separate PIE roots might, under the same probabilistic scrutiny, turn out to be artifacts of the same kind of taxonomic multiplication? This paper has not attempted to survey PIE reconstruction for other candidates. But the *$h_1$és- case, having been raised before, met with a narrow formal objection, and then left undeveloped for decades (Irslinger 2018, reporting Dunkel 2014), suggests that the field’s working assumption — that paradigmatic distinctions in the standard reconstruction reflect genuine historical distinctness unless conclusively shown otherwise — may itself merit reexamination as a default. The probabilistic method developed here offers one way of conducting that reexamination on a case-by-case basis, without requiring a wholesale revision of PIE reconstruction to get started.

Section 10. Conclusion

This paper has argued that the paradigmatic separation of *$h_1$és- into a copula and a third-person pronoun is not supported by the phonemic evidence. Treating PIE as a closed system for the relevant categories — pronouns and copulas being among the vocabulary least susceptible to borrowing — the probability of the observed three-phoneme identity arising by chance is approximately 1/27,000 under every laryngealist reconstruction surveyed, and remains overwhelming even under the most conservative non-laryngealist reconstruction available. This result does not depend on resolving any of the disputes that currently divide reconstructions of PIE; it holds across all of them.

The identity is not confined to the reconstructed proto-form. It is directly attested within the PIE oblique pronoun paradigm itself, and it persists, independently, across six daughter branches separated by thousands of years of divergent development — including Hittite, the most archaic attested witness, where the relevant laryngeal is read directly from the cuneiform record rather than reconstructed. It resurfaces again, independently of the inherited PIE pattern, in the medieval Ibero-Romance development of Spanish ser and estar. No single piece of this evidence would be decisive alone; together, they leave the hypothesis of coincidence without a credible mechanism.

The paper has also shown that this specific proposal is not without precedent. A pronominal origin for *$h_1$és- was raised before, on functional grounds resembling those developed here, and met with a formal objection turning on the morphological segmentation of the oblique pronoun paradigm. That objection, properly understood, addresses a question about morpheme boundaries, not about the phonemic sequence the probabilistic method actually compares. The two questions are independent, and resolving the present argument does not require resolving the segmentation dispute on which the earlier exchange turned.

If the argument is accepted, its implications extend beyond the single case examined here. The method developed in Sections 3 and 4 — treating relatedness in a closed system as a probabilistic question rather than an unresolvable one — offers a way of reopening other paradigmatic distinctions in PIE reconstruction that have been maintained by convention rather than by demonstrated necessity. And the typological evidence considered in Section 9 raises the further possibility that the separation in question may never have been salient to speakers of PIE at all, but is instead an artifact of the distance from which the comparative method is necessarily obliged to view a language with no surviving native speakers and no earlier written stage. Where the comparative method reaches its limit, probability, applied carefully and conservatively, can still speak.

References

Beekes, Robert S.P. 1995. *Comparative Indo-European Linguistics: An Introduction*. Amsterdam: John Benjamins.

Cooper, Adam. 2009. Similarity avoidance in the Proto-Indo-European root. *University of Pennsylvania Working Papers in Linguistics* 15: 55–64.

Corbeau, Alain Louis. 2013. *Phonological Root Structure Constraints in Proto-Indo-European*. ResMA Thesis, Leiden University.

Dunkel, George E. 2014. *Lexikon der indogermanischen Partikeln und Pronominalstämme*. 2 vols. Heidelberg: Universitätsverlag Winter.

Fortson, Benjamin W. IV. 2004. *Indo-European Language and Culture: An Introduction*. Oxford: Blackwell.

Hacking, Ian. 2001. *An Introduction to Probability and Inductive Logic*. Cambridge: Cambridge University Press.

Haspelmath, Martin, and Uri Tadmor (eds.). 2009. *Loanwords in the World's Languages: A Comparative Handbook*. Berlin: De Gruyter Mouton.

Irslinger, Britta. 2018. More tales of two copulas: The copula systems of Western European languages from a typological and diachronic perspective. In *Diachrony and Suppletion*, eds. Ronald Kim, Petr Zemanek, and Dita Frantikova. Hamburg: Baar.

Hartmann, Frederik. 2021. The phonetic value of the Proto-Indo-European laryngeals: A computational study using deep neural networks. *Indo-European Linguistics* 9: 26–84.

Iverson, Gregory K., and Joseph C. Salmons. 1992. The phonology of the Proto-Indo-European root structure constraints. *Lingua* 87: 293–320.

Katz, Aya. 1996. *Cyclical Grammaticalization and the Cognitive Link Between Pronoun and Copula*. PhD dissertation, Rice University. Available at: https://hdl.handle.net/1911/16974

Katz, Aya. 2001. Lexicalization and opacity. *LACUS Forum* 27: 113–121.

Meier-Brügger, Michael, Matthias Fritz, and Manfred Mayrhofer. 2010. *Indogermanische Sprachwissenschaft*. 9th ed. Berlin: De Gruyter.

Pokorny, Julius. 1959. *Indogermanisches etymologisches Wörterbuch*. Bern: Francke.

Rix, Helmut et al. 2001. *Lexikon der indogermanischen Verben (LIV)*. 2nd ed. Wiesbaden: Reichert.

Szemerényi, Oswald. 1980. *Introduction to Indo-European Linguistics*. Oxford: Oxford University Press.